Dated CM class-number bounds

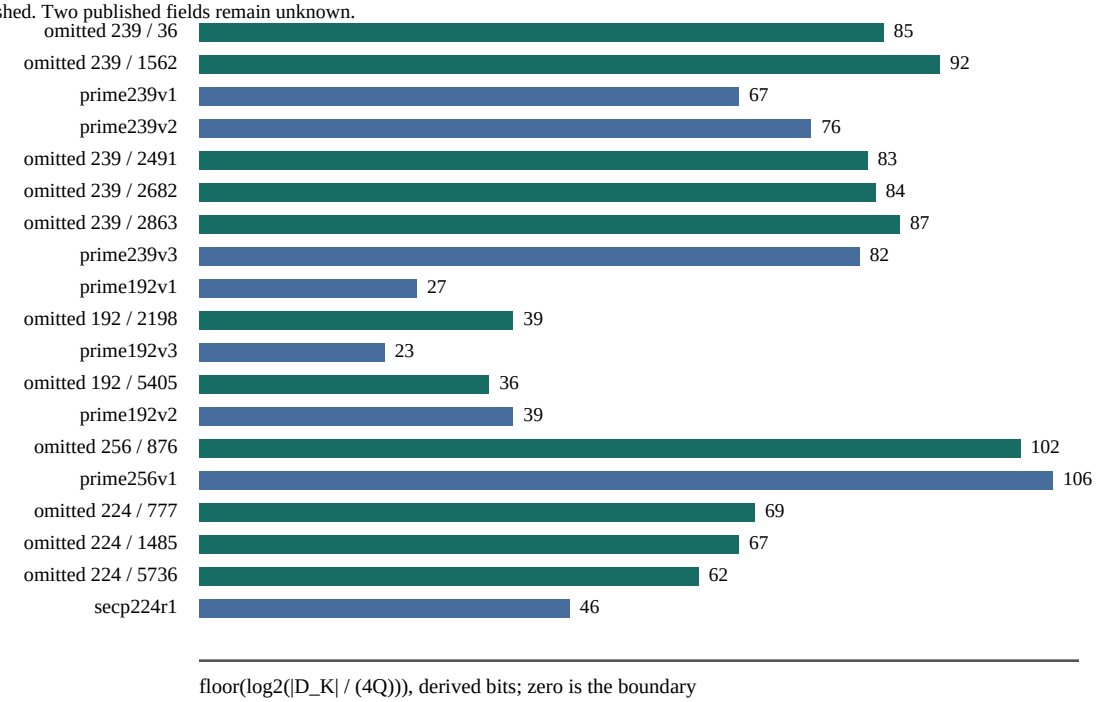
28 October 2026 UTC | T-5518 | observed exact arithmetic; historical selector unresolved

All eleven omissions clear both fixed bounds. Direct certificates prove h(K) >= 16,777,216 for eleven omitted and eight published curves. Published P-384 and P-521 remain UNKNOWN_D_K for this method; every one of the 21 identities is retained.

Dated criteria. Baier 2002 section2.3.5 and ETSI TS 102 176-1 V1.2.1 (July 2005),6.1.2.3/D.3, require h(K) >= 200. RFC 5639 (March 2010),2.1 item3, requires h(K) > 10^7. All refer to the maximal CM order. These later sources do not establish 1997 NIST adoption.

Certificate inequality margin for all 19 certified fields

Teal: omitted; blue: published. Two published fields remain unknown.



Certificate. Select the first 24 split primes q_i, Q=product(q_i). If 4Q < |D_K|, the 2^24 subset ideals represent distinct classes: equality would produce a principal integral ideal of squarefree norm 1 < N <= Q. A rational generator has square norm; a nonrational generator has norm >= |D_K|/4. Both are impossible. Full proof: CERTIFICATE-LEMMA.md.

Inference. At these dated bounds the predicate rejects zero omissions, so contributes no exclusions to a two-rule model. The class map onto the maximal-order class group is surjective; this lower bound also applies to the actual CM order. No exact class number, fitted threshold, historical adoption, or general curve-security conclusion is claimed.

Sources: d-nb.info/964515040/34; ETSI TS 102 176-1 V1.2.1 (2005-07); rfc-editor.org/rfc/rfc5639.html. Full URLs and source boundaries are in report.md and SOURCE-AUDIT.json. The 19 new certificates use no external low-class-number completeness theorem or GRH.

Complete cohort and retained execution

Curve / block	Label	Status / lower bound
omitted 239 / 36	omitted	>=16,777,216
omitted 239 / 1562	omitted	>=16,777,216
prime239v1	published	>=16,777,216
prime239v2	published	>=16,777,216
omitted 239 / 2491	omitted	>=16,777,216
omitted 239 / 2682	omitted	>=16,777,216
omitted 239 / 2863	omitted	>=16,777,216
prime239v3	published	>=16,777,216
prime192v1	published	>=16,777,216
omitted 192 / 2198	omitted	>=16,777,216
prime192v3	published	>=16,777,216
omitted 192 / 5405	omitted	>=16,777,216
prime192v2	published	>=16,777,216
secp521r1	published	UNKNOWN D_K
omitted 256 / 876	omitted	>=16,777,216
prime256v1	published	>=16,777,216
secp384r1	published	UNKNOWN D_K
omitted 224 / 777	omitted	>=16,777,216
omitted 224 / 1485	omitted	>=16,777,216
omitted 224 / 5736	omitted	>=16,777,216
secp224r1	published	>=16,777,216

Method and controls. Python 3.13.1 trial primes/Euler symbols and PARI 2.17.3 nextprime/Kronecker agree on every record. GP proves source-factor primality. Delta_pi, complete products and fundamental D_K relations check. Positive D=-23/-15, noncertificate D=-7, nonfundamental D=-75,A/A and both fail-closed guards pass. Exact models,orders,seeds,UIDs,prime lists and products: cohort.csv and version2 RESULT.json.

Amendment retained. Version 1 gave 17 certificates / 4 unknowns because the separately archived P-256 pair was omitted from input assembly. Version 2 adds only those two original source records; unchanged producer/audit bytes and selection rule. Each protocol was frozen and committed before its affected witnesses. Across both versions: 4 full-cohort evaluations completed, 0 unexpected failures/timeouts/retries, 4 deliberate guard failures retained. Both unified exports validate. Shared-desktop clocks are diagnostic only.

Conductor clarification. Both P-256 records have f_pi=1, hence f_E=1. The factor 3 in published P-256 D_K is ramification, not its conductor. A large Frobenius-order class number alone would not certify a large maximal-order class number.

Limits and next evidence. No original visitation log or 1997 use is established. Different thresholds, operational failures or other CM predicates remain outside this test; a new source-defined rule needs a prospective test. P-384/P-521 unknowns do not change the all-eleven result. Existing signed-action/SEA graphs are unchanged.

Protocol hashes: v1 1fa24263792c84ebf0040ffb7e1b0d03bdb0fe7d21040b5be976a5ca05e53c7e; v2 4371127490749adf5413200c7f60710c39347039b7ad179573ec7461438fecb4. Freeze commits72c533ee10/9c3872cc7f; raw commits542375f6b3/c58405ea3f. PR2602; report.md contains exact paths,commands and provenance.